Inference on Coefficients: Hypothesis Testing

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These lecture notes were originally developed by Dr. Kris Pruitt.

The purpose of this lesson is to demonstrate computational and mathematical methods for completing hypothesis tests for the slope parameter of a simple linear regression model. Specifically, we demonstrate the following six steps of any hypothesis test.

- Define the null and alternate hypotheses
- Choose a level of significance
- Compute a test statistic
- Construct a null distribution
- Calculate a p-value
- Decide whether to reject null hypothesis

The construction of the null distribution is the primary step impacted by the choice of computational versus mathematical approaches.

Computational Hypothesis Test

In the previous lesson, we derived a point estimate for the slope parameter of the following simple linear regression model.

$$y = \beta_0 + \beta_1 x + \epsilon$$

When fitting a regression model using sample data, the slope is a statistic intended to estimate the true slope in the population. Thus, we are estimating a population parameter β_1. Just as with any other population parameter, we can conduct hypothesis tests on the slope. The most common hypotheses are:

$$H_0: \ \beta_1 = 0$$
$$H_A: \ \beta_1 \neq 0$$

Let's think carefully about the meaning of the null hypothesis. If the slope parameter is equal to zero, then there is *no association* between the predictor variable and the response. Visually, a slope of zero implies a flat (horizontal) regression line. A flat regression line suggests that the response variable has the same average value regardless of the predictor value. By contrast, if the slope is not zero, then there is *some association* between the predictor and the response.

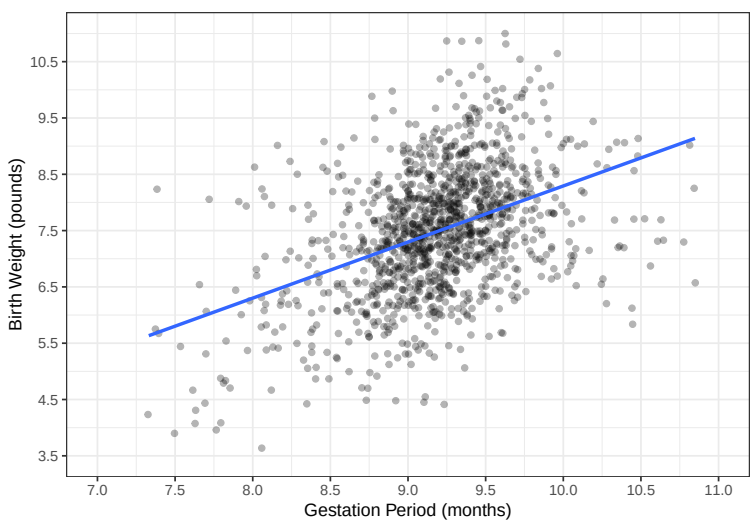
As an example, we will use a data set regarding the gestation period of babies from the early 1960s. As part of the Child Health and Development Studies (CHDS), this research investigated the impact of the mother's smoking habits on the health of the baby. Load the required packages and import the `Gestation` data frame.

```r
# load packages
library(mosaicData)
```

```r
# import data
data(Gestation)
```

Suppose we are interested in the association between gestation period (months) and birth weight (pounds). We'll begin by wrangling the data and visualizing the association in a scatter plot.

```r
# wrangle data
births <- Gestation %>%
  transmute(
    months = gestation / 30.42,
    weight = wt / 16
  ) %>%
  na.omit() %>%
  filter(months > 7, months < 11)

# plot gestation versus weight
ggplot(data = births, aes(x = months, y = weight)) +
  geom_point(position = "jitter", alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  scale_x_continuous(limits = c(7, 11), breaks = seq(7, 11, 0.5)) +
  scale_y_continuous(limits = c(3.5, 11), breaks = seq(3.5, 11, 1)) +
  labs(
    x = "Gestation Period (months)",
    y = "Birth Weight (pounds)"
  ) +
  theme_bw()
```



There does appear to be a weak, increasing, linear association between gestation period and birth weight. This makes intuitive sense, because the longer a baby has to grow in the womb, the heavier we would expect it to be upon birth, on average. We can fit a simple linear regression model of this association using the `lm()` function.

```r
# fit regression model
model <- lm(weight ~ months, data = births)

# display model statistics
coefficients(summary(model))
```

```
##               Estimate Std. Error   t value     Pr(>|t|)
```

```
## (Intercept) -1.6673281 0.56027715 -2.975899 2.979145e-03
## months       0.9959822 0.06092158 16.348594 1.881806e-54
```

```r
model %>%
  summary() %>%
  coefficients()
```

```
##               Estimate Std. Error   t value     Pr(>|t|)
## (Intercept) -1.6673281 0.56027715 -2.975899 2.979145e-03
## months       0.9959822 0.06092158 16.348594 1.881806e-54
```

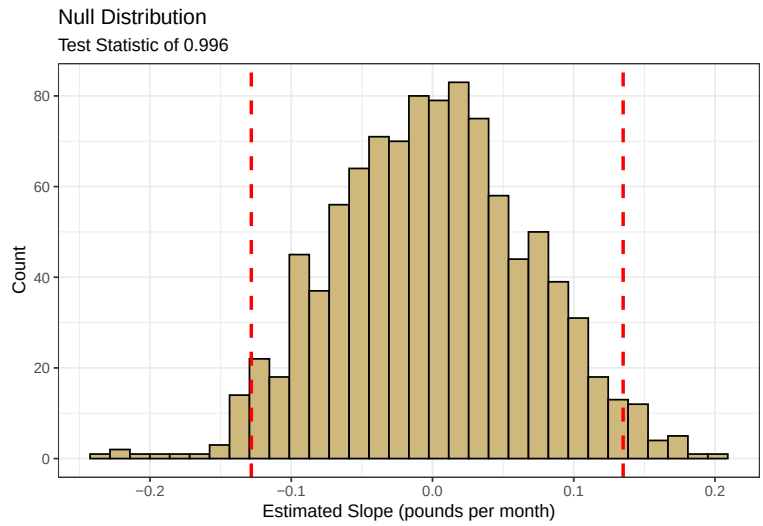
We have a point estimate for the slope of 0.996. In other words, our best estimate for growth rate is about 1 pound per month. However, we may have observed this nonzero sample slope purely due to chance. Perhaps we just obtained an odd sample of observations and the population slope is actually zero. We can evaluate this concern with a hypothesis test.

Returning to the hypotheses, we will define β_1 as the true slope parameter in units of pounds per month. Our null hypothesis suggests this slope is equal to zero, while the alternate claims it is not equal to zero. We'll choose a significance level of $\alpha = 0.05$ and we already know from our model summary that the test statistic is $\hat{\beta}_1 = 0.996$. Clearly the test statistic is non-zero, but we must compare this value to a null distribution in order to assess how significantly it differs from zero.

We can construct a null distribution using a method known as randomization. If we hold the predictor variable column static and randomly shuffle the values in the response variable column, then we are simulating *no association* between the two variables. If a certain gestation period (months) can be assigned any birth weight (pounds) within the range of the sample, then there is no clear relationship between the two. Let's conduct this randomization and generate a null distribution.

```r
# initiate empty vector
results <- data.frame(slope = rep(NA, 1000))

# repeat randomization process 1000 times and save results
for (i in 1:1000) {
  set.seed(i)
  shuffle <- births %>%
    transmute(months, weight,
      Shuffle_wt = sample(births$weight, replace = FALSE)
    )

  remodel <- lm(Shuffle_wt ~ months, data = shuffle)
  results$slope[i] <- coefficients(summary(remodel))[2, 1]
}

# compute rejection thresholds
threshold1 <- as.numeric(quantile(results$slope, 0.025))
threshold2 <- as.numeric(quantile(results$slope, 0.975))

# plot null distribution
ggplot(data = results, aes(x = slope)) +
  geom_histogram(color = "black", fill = "#CFB87C", bins = 32) +
  geom_vline(
    xintercept = threshold1, color = "red",
    linewidth = 1, linetype = "dashed"
  ) +
  geom_vline(
    xintercept = threshold2, color = "red",
    linewidth = 1, linetype = "dashed"
```

```r
  ) +
  labs(
    title = "Null Distribution",
    subtitle = "Test Statistic of 0.996",
    x = "Estimated Slope (pounds per month)",
    y = "Count"
  ) +
  theme_bw()
```



Our computational null distribution is centered on zero, as we would expect. Based on the rejection thresholds, we would reject the null hypothesis for any observed slope beyond about ± 0.15. Since our observed slope is nearly 1 with a p-value of effectively zero, we clearly reject the null hypothesis. There is significant statistical evidence to suggest that the true slope parameter is *not* zero. In other words, there *is* some association between gestation period and birth weight.

Mathematical Hypothesis Test

We can also apply mathematical approaches for hypothesis tests on the slope. Previously, we used randomization to generate a null distribution. Now we will use the results of the Central Limit Theorem (CLT), to assume a Normal null distribution. Our hypotheses remain the same.

$$H_0: \ \beta_1 = 0$$
$$H_A: \ \beta_1 \neq 0$$

As before, we define β_1 as the true slope parameter in units of pounds per month. Our null hypothesis suggests this slope is equal to zero, while the alternate claims it is not equal to zero. We'll choose a significance level of $\alpha = 0.05$ and the test statistic is $\hat{\beta}_1 = 0.996$.

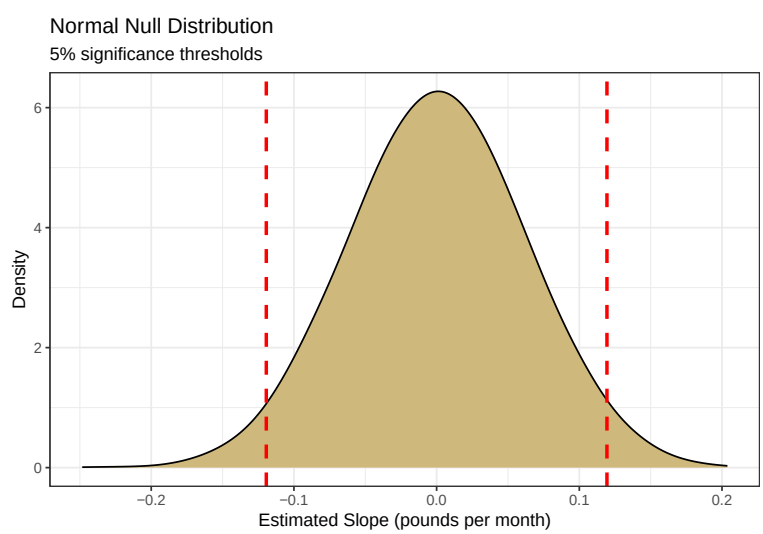
Rather than generating a randomized null distribution, we assume it is Normal with a fixed mean and standard deviation. The CLT supports this assumption as long as our sample is relatively **large** and **random**. Traditionally, a sample size of 30 observations is considered "large enough" but we are seldom constrained to so little data in this modern age. Clearly, our sample size of 1,216 is sufficiently large. To be considered random, our sample observations must be independent and identically distributed (iid). In this context, independence means the gestation period and birth weight of one child has no association with that of another child. Identical distribution means that there were no structural changes to the parameters of the population during the sampling process. As long as the CHDS conducted their sampling in a truly random manner, these assumptions should be satisfied.

Once we have assumed Normality, the question becomes what mean and standard deviation to employ for the null distribution. Given we are constructing a *null* distribution, the mean should be aligned with the *null* value. In other words, a slope equal to zero. Furthermore, we assume the null distribution has a standard deviation equal to the standard error of our test statistic. This value is determined by the following formula:

$$SE_{\hat{\beta}_1} = \sqrt{\frac{1}{n-2}\frac{\sum(y_i - \hat{y}_i)^2}{\sum(x_i - \bar{x})^2}}$$

The standard error indicates the expected variability in the estimated slope from one sample to the next. Given the complexity of the standard error formula, we typically rely on software to complete the calculations. In fact, this value is provided in the summary output of our model and is equal to 0.609. We visualize the null distribution as follows:

```r
# simulate Normal distribution
set.seed(303)
null <- data.frame(slope = rnorm(n = 10000, mean = 0, sd = 0.0609))

# plot null distribution
ggplot(data = null, aes(x = slope)) +
  geom_density(color = "black", fill = "#CFB87C", adjust = 2) +
  geom_vline(
    xintercept = qnorm(p = 0.025, mean = 0, sd = 0.0609),
    color = "red", linewidth = 1, linetype = "dashed"
  ) +
  geom_vline(
    xintercept = qnorm(p = 0.975, mean = 0, sd = 0.0609),
    color = "red", linewidth = 1, linetype = "dashed"
  ) +
  labs(
    title = "Normal Null Distribution",
    subtitle = "5% significance thresholds",
    x = "Estimated Slope (pounds per month)",
    y = "Density"
  ) +
  theme_bw()
```



Based on the rejection thresholds, we would reject the null hypothesis for any observed slope beyond about

± 0.12. Since our observed slope (test statistic) is nearly 1, we clearly reject the null hypothesis. Another approach is to compute the p-value and compare it to the significance level.

```r
# compute p-value
2 * pnorm(q = 0.996, mean = 0, sd = 0.0609, lower.tail = FALSE)
```

```
## [1] 4.028202e-60
```

The p-value is effectively zero and clearly less than $\alpha = 0.05$. Thus, we reject the null hypothesis. There is significant statistical evidence to suggest that the true slope parameter is *not* zero. In other words, there *is* some association between gestation period and birth weight. How do these results compare to our randomized hypothesis test?